

Census Income Classification & Customer Segmentation

*A Machine Learning Analysis for Retail
Marketing*

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May 6, 2026

Executive Summary

This project addresses two related retail marketing questions: how to identify individuals likely to earn more than \$50,000 per year, and how to group population into distinct, actionable customer segments. The analysis uses U.S. Census Bureau survey data with 199,523 records and 40 variables which entail the demographics, employment, household, and financial information of individuals.

Task 1, which was about predicting an income of an individual, framed income prediction as an imbalance binary classification as only 6.2% of records in the data had an income of over \$50,000 per year. Task 1 showed how Histogram-based Gradient Boosting Classification Tree produces the strongest classification result after threshold tuning.

Task 2 focused on segmenting the population for marketing purposes. The analysis first identified that one segment was primarily composed of children, who are not appropriate direct marketing targets and have limited purchasing power. To create more actionable customer groups, children were excluded and the segmentation was rerun on the adult population. Using K-Prototypes, which is suitable for mixed numeric and categorical data, the analysis identified five distinct adult segments with different demographic, employment, and financial profiles.

Introduction

Business Context

A retail business client wants to reach the right people with the right message to be intentional about their marketing. This report presents a data-driven approach to accomplish the two tasks. The first task builds and evaluates a classification model to predict whether an individual earns more than \$50,000 per year. The second task builds a segmentation model to divide the population into interpretable marketing groups. This report contains the data, methodology, findings, and recommendations from both analyses.

Data

The dataset is from the 1994 and 1995 Current Population Survey (CPS), conducted annually by the U.S. Census Bureau. The CPS is a stratified random sample of U.S. households designed to be representative of the civilian non-institutional population. The dataset contains 199,523 records and 42 columns. There are 40 demographic and employment variables, one sampling weight, and one income label.

The 1994 and 1995 median incomes were \$33,178 and \$34,076 respectively which reflects the economic conditions of the mid-1990s.

Target Variable

The target variable is the *label* column, which takes two values: **50000+**, indicating income greater than \$50,000, and **-50000**, indicating income of \$50,000 or less.

The class distribution is heavily imbalanced as 93.8% (187,141 rows) of the records fall in the lower-income class, and only 6.2% (12,382 rows) in the higher-income class. This reflects the genuine rarity of high earners in the mid-1990s U.S. population.

Population Weight

The *weight* column represents how many similar people in the broader population each record represents. For instance, a record with a weight of 10,000 represents more people than a record with a weight of 100.

The population weights helps the analysis better reflect the true population distribution rather than treating every survey response as equally representative.

Variable Overview

The 40 predictive variables cover six broad areas: demographics (age, sex, race, citizenship, Hispanic origin, country of birth), education (level attained and current enrollment), employment (occupation, industry, class of worker, employment status, hours and weeks worked, employer size), financial activity (capital gains, losses, dividends, business ownership, veterans benefits), household composition (marital status, family structure, tax filing status), and geographic background (country of birth, migration history). Not all variables were suitable for modeling. Features were excluded when they were redundant with other variables, too sparse to contribute reliable signal (several columns were 90–99% ‘Not in universe’, meaning the question simply did not apply to most individuals), or when their inclusion raised fairness concerns. The reasoning behind each exclusion is documented in the Feature Selection section.

Data Quality

The raw dataset required three cleaning steps. The process replaced “?” with proper missing values, stripped whitespace from categorical fields, and retained “Not in universe” as a valid category rather than treating it as missing. After cleaning, only the variable *country of birth self* had true missing values, affecting 1.7% of records.

Methods

Exploratory Data Analysis

A systematic exploration of the dataset was conducted before any modeling to identify which variables carry the strongest signal and to surface patterns with direct business implications. The key findings are presented below.

Education Shows the Clearest Income Gradient

Education is the strongest single demographic predictor of income. Professional degree holders (MD, JD, DDS) had a 54.0% income rate, which means the percentage of individuals within a given group who earn more than \$50,000 a year. The income rate of professional degree holders is nearly nine times the 6.2% population average.

Education Level	Income >\$50K Rate	Share of Population
Professional degree (MD, JD, DDS)	54.0%	0.9%
Doctorate degree	52.0%	0.6%
Master’s degree	31.2%	3.3%
Bachelor’s degree	19.7%	10.0%
Associates degree (academic)	9.4%	2.2%
Some college, no degree	6.4%	13.9%
High school graduate	3.9%	24.3%
Less than high school	<1.6%	17.3%

Employment Status Is the Strongest Operationalizable Signal

Full-time workers had a 13.2% income rate versus 1.7% for those not in the labor force. Since 13.4% of the full dataset is not in the labor force, filtering this group before applying the classification model removes a large block of near-certain negatives at no cost.

Investment Activity Produces a Six-Fold Income Lift

Only 3.7% of individuals had any capital gains activity. However, among those who did, the income rate was 32.7% whereas those without capital gains consisted of 5.2%. This 6.3x lift makes investment activity one of the most discriminating signals in the dataset.

The Gender Income Gap Is Large and Age-Dependent

Males had a 10.2% income rate compared to 2.6% for females. However, this gap varies substantially by age, widening sharply through mid-career.

Age Group	Male Income Rate	Female Income Rate	Gap
18–25	1.1%	0.6%	0.5pp
26–35	10.7%	3.1%	7.6pp
36–45	21.5%	5.4%	16.1pp
46–55	27.2%	6.2%	21.0pp
56–65	17.8%	3.5%	14.3pp
65+	6.7%	1.7%	5.0pp

These patterns are consistent with Bureau of Labor Statistics research from this period. Sex alone is not a reliable targeting variable and it must be considered alongside age.

Self-Employment Is a Surprisingly Strong Signal

Incorporated self-employed individuals had the highest income rate of any class of worker at 34.7% which is more than three times the 10.2% rate for private-sector employees. This suggests that business ownership type, not just employment status, is a meaningful income predictor.

Tax Filing Status Is a Powerful Household Proxy

Joint tax filers with both parties under 65 had a 13.1% income rate, compared to 6.1% for single filers. Non-filers which consisted of 37.6% of the dataset had a 0.0% rate.

Naturalized Citizens Outperform Native-Born Individuals

Foreign-born naturalized citizens had a 9.9% income rate, above the native US-born rate of 6.3%. Foreign-born non-citizens had a 3.7% rate. Treating all foreign-born individuals as a single low-income group would miss this meaningful high-income sub-segment.

Potential Multicollinearity

Potential collinearity was identified between family members under 18, age, and marital status, which was addressed in logistic regression via L2 regularization but is inconsequential for tree-based models.

Feature Engineering

Net Investment Income (`overall_gains_signed_log`)

Capital gains, capital losses, and dividends were combined into a single net investment income figure. The composite was then log-transformed using a signed log approach. Raw values reached nearly \$200,000, causing numerical overflow in logistic regression. The transformation compresses the extreme tail while preserving sign and relative ordering, bounding values between approximately -8 and +12.

Hourly Wage Indicator (`has_hourly_wage`)

Wage per hour is zero for salaried workers and non-zero for hourly workers. A binary indicator was created rather than using the raw value, which would introduce a heavily skewed distribution.

Hispanic Origin Indicator (`hispanic_origin_indicator`)

Collapsed from nine categories to a binary indicator where 1 would indicate if a specific Hispanic origin is recorded, 0 otherwise.

US-Born Indicator (`is_us_born`) - Segmentation Model Only

Country of birth self was collapsed to a binary indicator for the segmentation model, distinguishing US-born from foreign-born individuals more cleanly than the full categorical variable.

Feature Selection

Several features were dropped to improve model performance and interpretability. Two columns, detailed industry recode and detailed occupation recode, were excluded as they are more specific versions of the major-level codes and add noise without additional signal. A number of sparse columns were also removed: four migration code columns averaging 50% missing values, region and state of previous residence at 92% Not in universe, member of a labor union at 90%, reason for unemployment at 97%, and live in this house 1 year ago at 50%. Country of birth for father and mother were excluded on fairness grounds, as parental birthplace is several steps removed from individual income and could introduce bias against immigrant populations. The fill inc questionnaire for veteran's admin column was 99% Not in universe and carries no predictive signal. Finally, the weight column was used as `sample_weight` during training rather than as a feature, as including it as a predictor would constitute data leakage of the survey design.

Task 1 - Income Classification

Model Selection and Rationale

Logistic regression, random forest, and HistGradient Boosting were trained and compared to ensure the best approach was selected rather than relying on a single method.

Logistic Regression serves as the interpretable baseline as it is straightforward to audit and explains its predictions in terms of which variables matter most. Random Forest and HistGradientBoosting are both more powerful tree-based approaches that capture complex relationships between variables that a linear model would miss. The key difference between them is that Random Forest builds many trees independently and combines their votes, while HistGradientBoosting builds trees sequentially with each one correcting the errors of the last. HistGradientBoosting also includes automatic early stopping, which prevents it from over-learning the training data.

Handling Class Imbalance

Because only 6.2% of individuals in the dataset earn above \$50,000, a model trained without any adjustment would be biased toward predicting the majority class. To address this, all three models were configured to treat high-income individuals as proportionally more important during training, ensuring the model learns to identify them rather than defaulting to the easier majority prediction.

Additionally, for each model, the probability cutoff to identify high income was adjusted to find the value that best balances catching high-income individuals against avoiding false alarms. This tuning step was performed using the validation data and directly affects how the model performs in practice.

Population-Representative Training and Evaluation

Each record in the census dataset represents a different number of real people, as indicated by the weight column. A record representing 18,000 people in the actual population should have more influence on the model than one representing 38 people. All three models were trained and evaluated with this weighting applied, so the reported performance figures reflect how the model would be expected to perform across the real U.S. population.

Task 2 - Customer Segmentation

Adult Filter

Initial segmentation on the full dataset produced one cluster composed entirely of minors (average age 7.8, 0% income over \$50,000, 100% not in the labour force). Children have limited purchasing power, cannot be marketed to directly, and their presence distorted the remaining adult clusters. They were therefore excluded from the segmentation. Filtering to age ≥ 18 removed 55,992 records (28.1%) and raised the income rate from 6.2% to 8.7%.

Why K-Prototypes Over K-Means

K-Means uses Euclidean distance which has limitations for categorical variables. However, 13 of 21 segmentation features are categorical. As a result, K-Prototypes which instead use Hamming distance for categoricals and Euclidean for numerics were utilized.

Choosing k

The elbow method and silhouette score were evaluated for $k=2$ to 8 (K-Means) and $k=2$ to 7 (K-Prototypes). Five clusters were selected at the point where the cost reduction per additional cluster fell below 8%, indicating diminishing returns, while clusters remained sufficiently distinct to be interpretable for marketing purposes.

Results

Task 1 - Classification Model Performance

Because the high-income group represents only 6.2% of the dataset, accuracy alone is not a reliable measure of model quality. A model could predict everyone as earning less than \$50,000 and still be correct most of the time. Therefore, this report emphasizes precision, recall, F1, ROC AUC, and PR AUC. Precision measures how many people flagged as high-income actually are high-income, while recall measures how many true high-income individuals the model successfully identifies. F1 balances precision

and recall. ROC AUC and PR AUC evaluate how well the model ranks individuals by likelihood of earning more than \$50,000, with PR AUC being especially useful for imbalanced datasets.

Model Comparison

Model	Threshold	ROC AUC	PR AUC	Precision	Recall	F1
Logistic Regression	Default 0.50	0.942	0.574	0.275	0.898	0.421
Logistic Regression	Tuned	0.942	0.574	0.521	0.603	0.559
Random Forest	Default 0.50	0.945	0.600	0.290	0.887	0.438
Random Forest	Tuned	0.945	0.600	0.543	0.584	0.563
HistGradientBoosting	Default 0.50	0.950	0.645	0.308	0.897	0.459
HistGradientBoosting	Tuned	0.950	0.645	0.602	0.600	0.601

Selected Model: HistGradientBoosting (Tuned Threshold)

HistGradientBoosting was selected as the final model based on its performance across all metrics. It achieved the highest ROC AUC of 0.950, meaning it correctly ranks a high-income individual above a low-income individual 95% of the time. Its PR AUC of 0.645 is meaningfully higher than Logistic Regression (0.574), indicating it handles class imbalance more effectively. At the tuned threshold, it achieved an F1 of 0.601, which is the highest of any configuration tested, identifying 6 out of every 10 flagged individuals as genuine high earners.

Task 2 - Segmentation Results

K-Prototypes with five clusters on the adult dataset (143,531 records) produced five distinct, interpretable segments. The income rate per cluster ranges from 0.8% to 28.5%, confirming that the lifestyle-based segmentation captures meaningful economic differences.

Cluster 0 - Retired and Non-working Women

25.9% of the adult population with income rate of 0.8%

Average age 56, near-zero weeks worked, 67% female, and 88% not in the labor force. This segment represents retired women and stay-at-home spouses. The 0.8% income rate reflects the near-total absence of employment income. Despite the low income rate, this is the largest adult segment and represents an active consumer group for health, home, and leisure products.

Cluster 1 - Foreign-born Working Adults

12.5% of the adult population with income rate of 4.9%

100% foreign-born, 56% Hispanic origin, 60% non-citizen, average age 42, and 29 weeks worked per year. The income rate of 4.9% is below the adult average of 8.7%. However, this segment is predominantly married with family responsibilities, making them consistent consumers of family-oriented and value-focused products. Multilingual and culturally tailored communications are likely to produce higher engagement than standard campaigns.

Cluster 2 - Self-employed and Small Business Owners

11.4% of the adult population with income rate of 8.8%

Defined by the highest self-employment rate of any cluster, average age 34, and 44 weeks worked per year. The income rate sits at the adult population average, but self-employed individuals at this age are often in the growth phase of a business with distinct purchasing needs for supplies, technology, and professional services. This segment is more responsive to small business-oriented retail and financial products.

Cluster 3 - Wealthy Investor Households

12.1% of the adult population with income rate of 28.5%

The highest-value segment for premium marketing. At 28.5%, the income rate is 3.3 times the adult population average of 8.7%. The segment is defined by the highest net investment income of any cluster, average age 52, 72% married, bachelor's degree most common, and 96% US-born. This combination of employment income, investment income, and established household structure makes this segment the most commercially attractive for premium retail, luxury products, and financial services.

Cluster 4 - Core Working Adults

38.1% of the adult population with income rate of 8.8%

This segment is the largest adult segment. Average age 39, 48 weeks worked per year which is the highest of any cluster, predominantly administrative and clerical occupations, private sector, and virtually all US-born. The income rate is at the population average, but the segment's size and employment consistency make it the most reliable target for high-volume, everyday retail.

Segment Summary

Cluster	Name	Size	Income >\$50K	vs. Adult Average
0	Retired / Non-working Women	25.9%	0.8%	0.09x
1	Foreign-born Working Adults	12.5%	4.9%	0.56x
2	Self-employed / Small Business	11.4%	8.8%	1.01x
3	Wealthy Investor Households	12.1%	28.5%	3.28x
4	Core Working Adults	38.1%	8.8%	1.01x

Cluster 3 stands out as the clear premium target, while Clusters 0 and 1 are better suited to value-oriented strategies.

Discussion

Key Findings and Recommendations

Education Is the Most Reliable Pre-Filter for Premium Targeting

Professional and doctorate degree holders exceed a 50% income rate despite representing under 1.5% of the population. We recommend using education level as a fast pre-filter before applying the full model as focusing on individuals with at minimum a bachelor's degree targets a population where one in five earns above the income threshold.

Investment Activity Is the Highest-Precision Signal Available

Individuals with any capital gains activity have a 32.7% income rate versus 5.2% for those without. Where investment account or brokerage data is available, we recommend using it as a fast, high-confidence identifier of premium prospects. This finding also explains why Cluster 3 separates so clearly from the rest.

Employment Status Enables Cost-Free Pre-Screening

We recommend filtering out individuals not in the labor force before running the full model. This filter can be applied at the prospect list stage before any outreach budget is committed, removing 13.4% of the prospect pool while eliminating near-certain negatives and materially improving campaign precision at no additional cost.

The Gender Gap Requires Age-Sex Interaction Targeting

At ages 18–25, the income gap between men and women is negligible. By ages 46–55, it reaches 21 percentage points. We recommend incorporating age-sex interaction into any targeting strategy rather than using either variable alone. Mid-career men aged 36–55 represent the highest-precision demographic slice for premium marketing, and messaging and product selection for this group should reflect their higher purchasing power relative to other demographic combinations.

Prioritize Cluster 3 for Premium Campaigns

Wealthy Investor Households at 28.5% income rate represent a 3.3x lift over the adult average. This is the highest-value segment for luxury retail, financial services, and premium product launches. As a result, we recommend any campaign with a limited budget to be concentrated here first. Given their age profile, investment activity, and household stability, this segment is likely to respond well to financial services partnerships, wealth management products, home improvement, and travel.

Cluster 4 Represents the Highest Volume Opportunity

Core Working Adults, as the largest segment at 38.1% of the adult population, represent the highest volume marketing opportunity. Their consistent year-round employment and family-age profile make them well suited to everyday essentials, convenience retail, and family-oriented products. While their income rate sits at the population average, their size and spending consistency make them a reliable foundation for high-volume campaigns.

Limitations and Future Work

Several limitations should be noted. The dataset is from 1994–1995, and income distributions, labour market participation rates, and demographic compositions have changed significantly since then, meaning the model should be retrained on more recent census data before production deployment. Due to computational constraints, model comparison was based on a single stratified 80/20 split rather than k-fold cross-validation. The large training set of 159,618 rows reduces the risk of a split that is not representative of the population, but cross-validation would provide more robust estimates. While weighted and unweighted metrics were consistent overall, a formal fairness audit evaluating model performance separately by race, sex, and citizenship is appropriate before commercial deployment given the demographic patterns observed during exploration. Finally, the K-Prototypes segmentation uses random initialization, and while the five-cluster structure was stable across multiple runs with the same seed, a full stability analysis across different seeds and subsamples is recommended before production use.

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